

Course: B. Tech. Branch: Information Technology Semester: V

Subject Code & Name: BTITPE503E Data Visualization

Max Marks: 60

Date:11/08/2023

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) Explain the Seven Stages of Data Visualization.	CO 1	6
B) Discuss Advantages of Data Visualization.	CO 1	6
C) Explain the difference between Data Visualization and Infographics.	CO 1	6
Q.2 Solve Any Two of the following.		12
A) Discuss the Classification of Digital data.	CO 2	6
B) Explain with neat example reading JSON data into a Pandas DataFrame.	CO 2	6
C) What is a Python data dictionary? Explain with an Example.	CO 2	6
Q. 3 Solve Any Two of the following.		12
A) Discuss Histogram with suitable example.	CO 3	6
B) Explain Box plot with neat example.	CO 3	6
C) Explain Scatter plot with suitable data. Also write a python script to draw scatter plot.	CO 3	6
Q.4 Solve Any Two of the following.		12
A) Explain Data Wrangling in Python.	CO 4	6
B) Explain Pandas Correlations.	CO 4	6
C) Explain data merging and subsetting Data Frames in Pandas.	CO 4	6
Q. 5 Solve Any Two of the following.		12
A) Discuss the comparison between matplotlib and seaborn libraries.	CO 5	6
B) Explain Plotly library for Interactive Data Visualization in Python.	CO 5	6
C) Explain spatial data visualization.	CO 5	6

*** End ***

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Winter Examination – 2022

Course: B. Tech. Branch :Information Technology Semester : V
Subject Code & Name: BTITPE503E & Data Visualization
Max Marks: 60 Date: 02/02/2023 Duration: 03 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) What is the Need of Data visualization? Discuss the Evolution of data Visualization?	L1/CO1	6
B) Discuss the Classification of Digital data?	L2/CO4	6
C) What is Histogram? What is required to plot a Histogram discuss with example?	L4/CO2	6
Q.2 Solve Any Two of the following.		12
A) Discuss the Advantageous of Data Visualization?	L4/CO2	6
B) Discuss the pandas Library in Python with features?	L2/CO4	6
C) Write a python program to read the xml file "Bookcatlog.xml" located in "c:/data" directory. Display the rows which have id value as "bk 103"?	L6/CO5	6
Q. 3 Solve Any Two of the following.		12
A) Explain Iteration & Combinations with example.	L1/CO1	6
B) Discuss the Advantageous and Drawbacks of Unstructured Data?	L2/CO3	6
C) Explain process of Data Cleaning in Data Wrangling?	L3/CO4	6
Q.4 Solve Any Two of the following.		12
A) Discuss what is Iris Data set? And how Seaborn is effective for visualization?	L1/CO1	6
B) Which category (Structured, Semi-structured or Unstructured) you are going to place the following data. 1) Word Document. 2) PowerPoint Presentation. 3) CSV Data. 4) Spreadsheets. 5) Web page data.	L5/CO2	6
C) Explain tree map with syntax & example.	L4/CO4	6
Q. 5 Solve Any Two of the following.		12
A) Discuss the types of data also give the sources of such data with examples?	L2/ CO2	6
B) Discuss the comparison between matplotlib, seaborn and plotly libraries.	L2/ CO4	6

C)	Describe steps involved in making plots. Explain plotting two or more lines on the same plot with an example.	L3/ CO3	6
	*** End ***		

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3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) Explain the difference between Data Visualization and Infographics.	CO 1	6
B) Explain the Seven Stages of Data Visualization.	CO 1	6
C) Discuss the Classification of Digital data.	CO 2	6
Q.2 Solve Any Two of the following.		12
A) Consider the excel data sheet named by pokemon_data.csv with following column heads: Name, Type 1, Type 2, HP, Attack, Defense, Sp. Atk, Sp. Def, Speed, Generation, Legendary. Write a jupyter notebook script to: a) Read first 10 rows b) Make changes to the data and c) Save data to xlsx format.	CO 2	6
B) Explain with neat example reading XML into a Pandas DataFrame.	CO 2	6
C) What is a Python data dictionary? Explain with an Example.	CO 2	6
Q. 3 Solve Any Two of the following.		12
A) Write a Python programming to create a pie chart of gold medal achievements of five most successful countries in 2016 Summer Olympics. Read the data from a csv file medal.csv. Sample data: country, gold_medal United States,46 Great Britain,27 China,26 Russia,19 Germany,17	CO 3	6

- B) Explain Box plot with neat example. CO 3 6
- C) Write a Python programming to display a horizontal bar chart of the popularity of programming Languages. CO 3 6
- Sample data:
- Programing languages: Java, Python, PHP, JavaScript, C#, C++
- Popularity: 22.2, 17.6, 8.8, 8, 7.7, 6.7

Q.4 Solve Any Two of the following. 12

- A) What is Exploratory Data Analysis? Explain the Steps Involved in EDA. CO 4 6
- B) Explain Data Wrangling in Python. CO 4 6
- C) Write a Pandas program to detect missing values of a given Data Frame. CO 4 6
- Display True or False.

Test Data:

	ord_no	purch_amt	ord_date	customer_id	salesman_id
0	70001.0	150.50	2012-10-05	3002	5002.0
1	NaN	270.65	2012-09-10	3001	5003.0
2	70002.0	65.26	NaN	3001	5001.0
3	70004.0	110.50	2012-08-17	3003	NaN
4	NaN	948.50	2012-09-10	3002	5002.0
5	70005.0	2400.60	2012-07-27	3001	5001.0
6	NaN	5760.00	2012-09-10	3001	5001.0
7	70010.0	1983.43	2012-10-10	3004	NaN
8	70003.0	2480.40	2012-10-10	3003	5003.0
9	70012.0	250.45	2012-06-27	3002	5002.0
10	NaN	75.29	2012-08-17	3001	5003.0
11	70013.0	3045.60	2012-04-25	3001	NaN

Q. 5 Solve Any Two of the following. 12

- A) Discuss the comparison between matplotlib and seaborn libraries. CO 5 6
- B) Explain Plotly library for Interactive Data Visualization in Python. CO 5 6
- C) Consider the NSE stock market bhavcopy with following column heads: CO 5 6
- Symbol, Series, Date1, Prev_close, Open_price, High_price, Low_price, Last_price, Close_price, Avg_price, Ttl_trd_qnty, Turnover_lacs, No_of_trades, Deliv_qty, Deliv_per. Write the python script for NSE stock market bhavcopy analysis to display the scripts(shares) that are closed greater than the previous close price and turnover is greater than Rs. 100 lacs and also delivery percentage is greater than 90%.

*** End ***

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Summer Examination- 2024

Course: B. Tech. Branch: Information Technology Semester : V
Subject Code & Name: BTITPE503E Data Visualization
Max Marks: 60 Date: 05/07/2024 Duration: 03 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. Assume suitable data wherever necessary and mention it clearly.

		(Level)	Marks
Q. 1	Solve Any Two of the following.		12
A)	Explain Seven Stages of Data Visualization.	L1	06
B)	Differentiate between Structured and Unstructured data.	L2	06
C)	Explain why data display requires planning. What are the different steps to form a path for the process of understanding data?	L3	06
Q.2	Solve Any Two of the following.		12
A)	What are the various data sources from where the data can be read into a python data frame?	L2	06
B)	Develop a python program to plot two or more lines with legends, different widths and colours.	L2	06
C)	Prepare a excel sheet of different things available in supermarket and write a python programme to display the particular items availability.	L3	06
Q. 3	Solve Any Two of the following.		12
A)	What is a Histogram plot? Explain in detail the different terms used with Histogram. What are the pros of histogram plots?	L2	06
B)	What is a Scatter Plot? Keeping in mind the correlation coefficient visualize the relationships with suitable examples.	L3	06
C)	Define Matplotlib. Explain the architecture of Matplotlib.	L2	06
Q.4	Solve Any Two of the following.		12
A)	Explain Data Structure in Panda with example.	L1	06
B)	What is Data Wrangling in python? Discuss the operations included in wrangling for faster decision making and efficient analysis.	L2	06
C)	Explain process coding and Explain tree map with syntax & example.	L2	06

Q. 5	Solve Any Two of the following.		12
A)	What is Plotly Python package? Write a pseudo code to plot a Sine wave using plotly.graph_objs module.	L3	06
B)	What is Seaborn for Data Visualization? Write the code to draw the Pie chart of Iris data set.	L3	06
C)	Give comparison between Matplotlib, Seaborn and Plotly.	L4	06

*****End*****